Notes: Brandt, Tombe, and Zhu (RED, 2013)

Factor Market Distortions Across Time, Space and Sectors in China

Contents

1 Big picture

- **Question.** How large are the TFP losses in China's non-agricultural economy from misallocation of labor and capital across provinces and between the state and non-state sectors, and how did those losses evolve from 1985 to 2007?
- **Setting.** The paper studies a static yearly allocation problem in China's non-agricultural economy, treating each province as having two sectors—state and non-state—and measuring wedges in labor, capital, and product markets.
- **Core challenge.** Raw dispersion in returns to labor or capital does not map cleanly into aggregate productivity losses. The paper therefore builds a model that turns province-sector wedges into a direct measure of TFP loss.
- **Main theoretical result.** With Cobb–Douglas production at the province-sector level and CES aggregation across sectors and provinces, the efficient allocation has a simple "TFP-share" form. The gap between actual and efficient TFP is a model-based measure of the cost of distortions.
- **Main empirical result.** Factor misallocation lowers aggregate non-agricultural TFP by about 20% on average over 1985–2007. Distortions declined until the mid-1990s and then rose again, with the reversal driven almost entirely by capital misallocation between the state and non-state sectors within provinces.
- **Main decomposition result.** Between-province distortions remain fairly constant over time and are driven mainly by labor-market wedges, while within-province distortions explain most of the time variation in aggregate losses.
- **Economic takeaway.** The big post-1997 problem is not that capital failed to move across provinces in general; it is that policy pushed too much capital toward the state sector, especially in lower-productivity regions.
- **Why this paper matters.** It is a benchmark paper on spatial and ownership misallocation in China. It connects wedges to aggregate TFP, decomposes where the losses come from, and provides a quantitative explanation for why reform initially reduced distortions but later policy choices reversed part of those gains.

2 Data and measurement (key objects)

2.1 Province-sector panel

- **Unit of observation.** Province i and sector $j \in \{s, n\}$ in year t, where s denotes the state sector and n the non-state sector.
- **Coverage.** The sample covers 27 of China's 31 mainland provinces over 1985–2007. Chongqing is merged with Sichuan before 1997, and Tibet, Hainan, and Hunan are excluded because of missing data.
- **Economic scope.** The paper studies only *non-agricultural* activity, so all employment, capital, and GDP objects are defined for the non-primary sector of the economy.
- **Core variables.** Province-sector employment, capital stock, nominal GDP, real GDP growth, wages, and fixed investment, together with province-level price deflators.

2.2 Employment construction

- **Official source.** The National Bureau of Statistics reports provincial employment by agriculture/non-agriculture and state/non-state status.
- **Main data problems.** Provincial totals do not add up to national employment, migrants are often recorded in the province of hukou rather than where they work, employed persons include some unemployed workers, and primary-sector employment is overstated relative to non-primary employment.
- **Corrections.** The paper uses census microdata for 1982, 1990, 1995, 2000, and 2005 to redistribute employment across provinces, correct for migration and labor-force composition, and then adjusts primary employment using estimates from Brandt and Zhu (2010).
- **State versus non-state.** From 1993 onward, shareholding corporations are counted as part of the state sector. Non-state employment is then measured residually from corrected non-agricultural employment minus state employment.

2.3 Capital stock construction

- **Method.** Capital stocks are built by perpetual inventory from province-level fixed investment data, split into non-primary/state and non-primary/non-state components.
- **Deflation.** Investment is deflated using province-level asset-price indices after 1993 and out-of-sample forecasts of asset deflators before 1993.
- **Depreciation and initial stocks.** The baseline depreciation rate is 7%, and initial capital stocks are backcast to 1978 using investment growth.
- **Privatization adjustment.** Provincial stocks are rescaled to match national state and non-state capital totals after adjusting for privatization of state-owned enterprises in the late 1990s.

2.4 GDP, deflators, and sectoral output

- **Provincial GDP.** The NBS reports nominal GDP and real GDP growth but not real GDP levels, so the paper reconstructs real non-agricultural GDP using nominal GDP, real growth rates, and 1990 spatial price levels from Brandt and Holz (2006).
- **State/non-state split.** Because GDP is not fully reported by ownership, the paper infers each sector's GDP share from its wage-bill share, following Brandt and Zhu (2010).
- **Validation.** In manufacturing during 1998–2007, the average actual state-sector share of value added is 0.53, while the wage-bill-based estimate is 0.52, which is reassuring for the broader non-agricultural application.

2.5 Unobservables, timing, and measured TFP

- **TFP object.** Province-sector TFP is A_{ij}, the residual in a Cobb–Douglas production function once real output, labor, and capital are measured.

- **Sectoral prices are not observed.** Province-level GDP deflators exist, but separate state/non-state output prices do not. The paper therefore infers sectoral relative prices from nominal value-added shares using the CES product-market structure.
- **Static timing.** The model is solved separately year by year, taking aggregate labor and capital as given in each year and asking whether their observed allocation across provinces and sectors is efficient.

2.6 Baseline parameters and empirical applications

- **Labor elasticity.** The baseline sets $a = 0.67$, following Hsieh and Klenow's practice of using US-style factor elasticities rather than measured Chinese labor shares, which are distorted by wedges.
- **Elasticities of substitution.** The paper sets $\phi = 0.67$ and $\sigma = 0.67$, implying substitution elasticities of 1.5 across sectors and across provinces.
- **Province weights.** Province weights ω_i are calibrated so that the product-market first-order condition holds on average over 1985–2007 when product-market distortions are absent:

$$\omega_i = \frac{1}{23} \sum_{t=1985}^{2007} \frac{P_i(t) Y_i(t)^\sigma}{\sum_{i'=1}^{m} P_{i'}(t) Y_{i'}(t)^\sigma}.$$

- **Applications.** The paper measures aggregate distortions over time, decomposes them into within- and between-province components, compares China's between-province distortions to US between-state distortions, and studies robustness to infrastructure, human capital, and industry composition.

3 Models and empirical specifications (all equations + interpretation)

3.1 Production function and aggregation

The province-sector production function is

$$Y_{ij} = A_{ij} L_{ij}^{a} K_{ij}^{1-a}, \qquad 0 < a < 1.$$

Provincial GDP is a CES aggregate of state and non-state output:

$$Y_i = \left(Y_{in}^{1-\phi} + Y_{is}^{1-\phi} \right)^{\frac{1}{1-\phi}},$$

and aggregate GDP is a CES aggregate of provincial GDPs:

$$Y = \left(\sum_{i=1}^{m} \omega_i Y_i^{1-\sigma} \right)^{\frac{1}{1-\sigma}}.$$

Let

$$l_{j|i} = \frac{L_{ij}}{L_i}, \qquad k_{j|i} = \frac{K_{ij}}{K_i}, \qquad l_i = \frac{L_i}{L}, \qquad k_i = \frac{K_i}{K},$$

where $L_i = L_{in} + L_{is}$, $K_i = K_{in} + K_{is}$, $L = \sum_i L_i$, and $K = \sum_i K_i$.

For any allocation, provincial and aggregate TFP are

$$A_i = \frac{\left(Y_{is}^{1-\phi} + Y_{in}^{1-\phi}\right)^{\frac{1}{1-\phi}}}{L_i^a K_i^{1-a}} = \left[\left(A_{is} l_{s|i}^a k_{s|i}^{1-a}\right)^{1-\phi} + \left(A_{in} l_{n|i}^a k_{n|i}^{1-a}\right)^{1-\phi}\right]^{\frac{1}{1-\phi}},$$

$$A = \frac{\left(\sum_{i=1}^m \omega_i Y_i^{1-\sigma}\right)^{\frac{1}{1-\sigma}}}{L^a K^{1-a}} = \left[\sum_{i=1}^m \omega_i \left(A_i l_i^a k_i^{1-a}\right)^{1-\sigma}\right]^{\frac{1}{1-\sigma}}.$$

- **Interpretation.** The paper measures distortions at a fairly aggregated level: between provinces and between ownership sectors inside provinces. Because outputs are imperfect substitutes across both dimensions, high-TFP locations do not absorb all factors even in the efficient allocation.
- **Why gross output is not the object here.** The focus is not on plant-level production-function estimation but on the efficient allocation of already-measured provincial factor supplies across province-sector cells.

3.2 Efficient allocation and TFP losses

The efficient allocation solves

$$\max_{\{L_{ij}, K_{ij}\}} Y$$

subject to the production and aggregation equations above and the aggregate resource constraints

$$\sum_{i,j} L_{ij} = L, \qquad \sum_{i,j} K_{ij} = K.$$

Proposition 1 shows that the efficient allocation has the form

$$\frac{L_{ij}}{L_i} = \frac{K_{ij}}{K_i} = \pi_{j|i}, \qquad \frac{L_i}{L} = \frac{K_i}{K} = \pi_i,$$

where

$$\pi_{j|i} = \left(\frac{A_{ij}}{A_i^*}\right)^{\frac{1-\phi}{\phi}} = \frac{A_{ij}^{\frac{1-\phi}{\phi}}}{A_{is}^{\frac{1-\phi}{\phi}} + A_{in}^{\frac{1-\phi}{\phi}}}, \qquad A_i^* = \left(A_{is}^{\frac{1-\phi}{\phi}} + A_{in}^{\frac{1-\phi}{\phi}}\right)^{\frac{\phi}{1-\phi}},$$

and

$$\pi_i = \frac{\omega_i^{1/\sigma} \left(A_i^*\right)^{\frac{1-\sigma}{\sigma}}}{\sum_{i'=1}^m \omega_{i'}^{1/\sigma} \left(A_{i'}^*\right)^{\frac{1-\sigma}{\sigma}}}.$$

Therefore, efficient aggregate TFP is

$$A^* = \left[\sum_{i=1}^m \omega_i^{1/\sigma} \left(A_i^*\right)^{\frac{1-\sigma}{\sigma}}\right]^{\frac{\sigma}{1-\sigma}}.$$

The TFP losses from distortions are then measured as

$$D = \ln\left(\frac{A^*}{A}\right), \qquad D_i = \ln\left(\frac{A_i^*}{A_i}\right).$$

- **Interpretation.** Efficient factor shares are literally "TFP shares" within provinces and across provinces. If a province-sector cell has higher TFP, it should attract a larger share of both labor and capital.
- **Why D is useful.** This gives a direct mapping from measured wedges to aggregate productivity losses, which is what simple return-dispersion statistics do not provide.

3.3 Competitive allocation with wedges

The paper introduces three wedges: a province-specific output wedge τ_i^y and province-sector-specific labor and capital wedges τ_{ij}^l and τ_{ij}^k.

At the top level, the stand-in firm that aggregates provincial output solves

$$\max_{\{Y_i\}} \left\{ P \left(\sum_{i=1}^{m} \omega_i Y_i^{1-\sigma} \right)^{\frac{1}{1-\sigma}} - \sum_{i=1}^{m} \tau_i^y P_i Y_i \right\},$$

which implies

$$\tau_i^y P_i = \omega_i P \left(\frac{Y_i}{Y} \right)^{-\sigma}.$$

At the province level, the aggregator solves

$$\max_{Y_{is}, Y_{in}} \left\{ P_i \left(Y_{is}^{1-\phi} + Y_{in}^{1-\phi} \right)^{\frac{1}{1-\phi}} - P_{is} Y_{is} - P_{in} Y_{in} \right\},$$

so sectoral output prices satisfy

$$P_{ij} = P_i \left(\frac{Y_{ij}}{Y_i} \right)^{-\phi}.$$

The province price index is

$$P_i = \left(P_{is}^{\frac{\phi-1}{\phi}} + P_{in}^{\frac{\phi-1}{\phi}} \right)^{\frac{\phi}{\phi-1}},$$

and the aggregate price index is

$$P = \left(\sum_{i=1}^{m} \omega_i^{1/\sigma} \hat{P}_i^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}, \qquad \hat{P}_i \equiv \tau_i^y P_i.$$

At the bottom level, the stand-in firm in province i and sector j solves

$$\max_{K_{ij}, L_{ij}} \left\{ P_{ij} A_{ij} L_{ij}^a K_{ij}^{1-a} - \tau_{ij}^l w L_{ij} - \tau_{ij}^k r K_{ij} \right\},$$

with first-order conditions

$$a P_{ij} A_{ij} L_{ij}^{a-1} K_{ij}^{1-a} = \tau_{ij}^l w, \qquad (1-a) P_{ij} A_{ij} L_{ij}^a K_{ij}^{-a} = \tau_{ij}^k r.$$

- **Interpretation.** Labor and capital wedges drive a gap between marginal products and common factor prices. The output wedge drives a gap between provincial marginal revenue and provincial marginal cost in the aggregate CES system.
- **Why this decomposition is useful.** It separates distortions in the factor markets from distortions in the spatial product market.

3.4 Competitive allocation in terms of wedge-adjusted TFP

Define composite wedges

$$\tilde{\tau}_{ij}^{l} = \tau_i^y \tau_{ij}^l, \qquad \tilde{\tau}_{ij}^{k} = \tau_i^y \tau_{ij}^k,$$

and wedge-adjusted TFP

$$\tilde{A}_{ij} = \frac{A_{ij}}{(\tilde{\tau}_{ij}^l)^a (\tilde{\tau}_{ij}^k)^{1-a}}, \qquad \tilde{A}_i = \left(\tilde{A}_{is}^{\frac{1-\phi}{\phi}} + \tilde{A}_{in}^{\frac{1-\phi}{\phi}}\right)^{\frac{\phi}{1-\phi}}.$$

The paper then defines $\tilde{\tau}_i^l$ and $\tilde{\tau}_i^k$ as TFP-weighted harmonic averages of the within-province sectoral wedges, and $\tilde{\tau}^l$ and $\tilde{\tau}^k$ as the analogous economy-wide averages across provinces.

Proposition 2 implies

$$l_{j|i} = \frac{\tilde{A}_{ij}^{\frac{1-\phi}{\phi}} (\tilde{\tau}_{ij}^l)^{-1}}{\tilde{A}_{is}^{\frac{1-\phi}{\phi}} (\tilde{\tau}_{is}^l)^{-1} + \tilde{A}_{in}^{\frac{1-\phi}{\phi}} (\tilde{\tau}_{in}^l)^{-1}}, \qquad k_{j|i} = \frac{\tilde{A}_{ij}^{\frac{1-\phi}{\phi}} (\tilde{\tau}_{ij}^k)^{-1}}{\tilde{A}_{is}^{\frac{1-\phi}{\phi}} (\tilde{\tau}_{is}^k)^{-1} + \tilde{A}_{in}^{\frac{1-\phi}{\phi}} (\tilde{\tau}_{in}^k)^{-1}},$$

$$l_i = \frac{\omega_i^{1/\sigma} \tilde{A}_i^{\frac{1-\sigma}{\sigma}} (\tilde{\tau}_i^l)^{-1}}{\sum_{i'=1}^m \omega_{i'}^{1/\sigma} \tilde{A}_{i'}^{\frac{1-\sigma}{\sigma}} (\tilde{\tau}_{i'}^l)^{-1}}, \qquad k_i = \frac{\omega_i^{1/\sigma} \tilde{A}_i^{\frac{1-\sigma}{\sigma}} (\tilde{\tau}_i^k)^{-1}}{\sum_{i'=1}^m \omega_{i'}^{1/\sigma} \tilde{A}_{i'}^{\frac{1-\sigma}{\sigma}} (\tilde{\tau}_{i'}^k)^{-1}}.$$

Provincial and aggregate TFP under distortions are then

$$A_i = \tilde{A}_i (\tilde{\tau}_i^l)^a (\tilde{\tau}_i^k)^{1-a}, \qquad A = \left[\sum_{i=1}^m \omega_i^{1/\sigma} \tilde{A}_i^{\frac{1-\sigma}{\sigma}}\right]^{\frac{\sigma}{1-\sigma}} (\tilde{\tau}^l)^a (\tilde{\tau}^k)^{1-a}.$$

- **Interpretation.** Allocation depends on wedge-adjusted TFP and inverse wedges. High factor wedges reduce the effective attractiveness of a province-sector cell.
- **Key insight.** Factor allocation depends on the *product* of the output wedge and factor wedges. Therefore, allocation alone cannot separate product-market and factor-market distortions.

3.5 Identification of wedges

Proposition 3 shows that the observed allocation of labor and capital together with provincial price levels identifies the wedges up to a scalar. In particular,

$$\tau_{ij}^l \propto \frac{P_{ij} Y_{ij}}{L_{ij}}, \qquad \tau_{ij}^k \propto \frac{P_{ij} Y_{ij}}{K_{ij}},$$

so labor and capital wedges are proportional to average value products. The province output wedge is pinned down by the top-level first-order condition,

$$\tau_i^y \propto P_i^{-1} \omega_i \left(\frac{Y_i}{Y}\right)^{-\sigma},$$

which uses relative provincial prices to separate output wedges from factor wedges.

- **Interpretation.** Once relative prices are observed, the model has enough information to recover the distortions that rationalize the actual allocation.
- **Why only up to scale?** A common proportional change in all labor wedges, or in all capital wedges, leaves allocation unchanged. Since only relative wedges matter, the paper normalizes the common components away.

3.6 Sectoral prices and measured TFP

Because sectoral output deflators are unavailable, the paper infers them from nominal value-added shares using the CES structure:

$$\frac{P_{ij}(t)}{P_i(t)} = \left(\frac{Y_{ij}^{\text{nom}}(t)}{Y_{is}^{\text{nom}}(t) + Y_{in}^{\text{nom}}(t)} \right)^{-\frac{\phi}{1-\phi}}.$$

Hence real value added by province and sector is

$$Y_{ij}(t) = \frac{Y_{ij}^{\text{nom}}(t)}{P_{ij}(t)} = \frac{Y_{ij}^{\text{nom}}(t)}{P_i(t)} \left(\frac{Y_{ij}^{\text{nom}}(t)}{Y_{is}^{\text{nom}}(t) + Y_{in}^{\text{nom}}(t)} \right)^{\frac{\phi}{1-\phi}}.$$

The paper then backs out province-sector TFP from

$$A_{ij} = \frac{Y_{ij}}{L_{ij}^{a} K_{ij}^{1-a}}.$$

- **Interpretation.** This is a Hsieh–Klenow style move: use equilibrium conditions to recover missing price information from nominal shares.
- **Why it matters.** Without sectoral prices, one could not measure state-sector and non-state-sector real output separately, and the entire decomposition would collapse.

3.7 Counterfactual decompositions

Let A^{nw} denote aggregate TFP when within-province distortions are eliminated and A^{nb} denote aggregate TFP when between-province distortions are eliminated. Then the paper defines

$$D_b = \ln\left(\frac{A^*}{A^{nw}}\right), \qquad D^w = \ln\left(\frac{A^*}{A^{nb}}\right),$$

as the between-province and within-province components of the overall distortion. Their contributions to actual TFP loss are

$$d_w = D - D_b = \ln\left(\frac{A^{nw}}{A}\right), \qquad d_b = D - D^w = \ln\left(\frac{A^{nb}}{A}\right).$$

The paper then runs further counterfactuals eliminating only within-province capital wedges, only within-province labor wedges, only between-province capital wedges, only between-province labor wedges, or only product-market wedges.

- **Interpretation.** These are not separate regressions but model-based counterfactual experiments. Each one asks how much TFP would rise if a specific set of wedges were equalized.
- **Why this is powerful.** It lets the authors say not only that distortions are large, but also whether they come from labor, capital, or product markets and whether they arise across provinces or across ownership sectors within provinces.

3.8 Robustness extensions

The main robustness exercise adds infrastructure capital. State-sector capital is split into infrastructure X_i and non-infrastructure capital K_{is}, and both sectors use infrastructure as an input:

$$Y_{ij} = A_{ij} L_{ij}^{a} K_{ij}^{b} X_i^{1-a-b}.$$

If the government chose infrastructure allocation optimally, the model implies

$$\frac{X_i}{K_i} = \frac{1-a-b}{1-a}, \qquad K_i = K_{is} + K_{in} + X_i.$$

The paper sets $b = 0.25$, implying an infrastructure elasticity of 0.08, to match the average infrastructure share in the data.

Two other robustness checks are important:

- **Human capital adjustment.** Replacing raw labor with human-capital-adjusted labor lowers the average distortion slightly—to just under 0.18 instead of 0.20—but leaves the main patterns unchanged.
- **Industry composition.** Using US labor shares to infer different state and non-state factor elasticities suggests that the state sector has actually become *more labor intensive*, not more capital intensive. So the post-1997 increase in capital misallocation is not an artifact of hidden sector-composition changes.

4 Identification and instruments (what makes the estimates credible)

4.1 What simpler approaches miss

- **Separate return dispersions are not enough.** Earlier work on China often documented high returns to capital in one place and low returns elsewhere. But those statistics do not tell us how much aggregate TFP is lost because they do not embed the full allocation problem.
- **Firm-level misallocation papers answer a different question.** Hsieh and Klenow focus on within-industry misallocation in manufacturing over a shorter, later period. This paper instead asks about distortions across provinces and across ownership sectors in the broader non-agricultural economy over 1985–2007.
- **Why that matters.** The policy institutions in China—hukou restrictions, local protectionism, directed credit, and state favoritism—operate strongly across space and ownership, so the relevant aggregation level is province by sector.

4.2 What identifies the model here

- **Observed allocation.** The actual shares of labor and capital across provinces and sectors identify the factor wedges up to scale.
- **Relative provincial prices.** Provincial price levels add exactly the missing information needed to separate output wedges from factor wedges.

- **Sectoral price inference.** Because sectoral deflators are not observed, the CES product-market structure converts nominal value-added shares into sectoral relative prices and thus real state/non-state output.
- **Counterfactual logic.** Once wedges are identified, the model itself defines the efficient allocation and the no-within or no-between counterfactuals. The decomposition is therefore internally consistent.

4.3 Why the empirical decomposition is credible

- **Conservative parameters.** The substitution elasticities are set at 1.5, which is lower than values often used in trade work, so the baseline does not mechanically overstate gains from reallocation.
- **Multiple data corrections.** Employment, capital, and GDP are all corrected for well-known problems in official Chinese data, especially migration, privatization, and spatial price differences.
- **Robustness.** The post-1997 rise in distortions survives alternative parameter values, infrastructure accounting, human-capital adjustment, and relaxing equal factor elasticities across sectors.

4.4 Relation to the literature

- **Relative to Hsieh and Klenow (2009).** This paper studies broader spatial and ownership distortions, not within-four-digit-industry distortions. It also finds a different post-1997 pattern: worsening distortions in the non-agricultural economy as a whole.
- **Relative to Song et al. (2011).** Song et al. emphasize wedges in returns to capital between state and non-state sectors, but they do not quantify aggregate TFP losses or integrate the spatial dimension.
- **Main contribution.** The paper turns misallocation in China from a descriptive claim about distorted returns into a quantitative decomposition of aggregate productivity losses across time, space, and sectors.

5 Tables and figures

5.1 Table 1 and Figure 1: TFP levels by province, sector, and region

Read:

- Table 1 shows that non-state TFP is much higher than state-sector TFP in virtually every province and every year, and the gap generally widens over time.
- At the regional level, the East has the highest aggregate TFP, rising from about 0.454 in 1985 to 1.971 in 2007, while the West rises from about 0.384 to only 1.034.
- Figure 1 shows the entire cross-province distribution of log TFP by year and sector. State-sector TFP is relatively flat over time, whereas non-state TFP rises steadily and its cross-province dispersion narrows somewhat.

Table 1
Total factor productivity for selected years by province and sector.

Province	Aggregate			State			Non-state		
	1985	1997	2007	1985	1997	2007	1985	1997	2007
Anhui	0.353	0.854	1.545	0.033	0.024	0.036	0.227	0.676	1.338
Beijing	0.574	1.047	1.976	0.304	0.397	0.153	0.052	0.152	1.221
Fujian	0.406	1.154	1.946	0.037	0.060	0.087	0.216	0.734	1.327
Gansu	0.317	0.622	1.035	0.090	0.153	0.158	0.107	0.203	0.525
Guangdong	0.362	0.969	1.925	0.041	0.074	0.077	0.177	0.550	1.363
Guangxi	0.364	0.699	1.198	0.099	0.064	0.076	0.124	0.370	0.789
Guizhou	0.237	0.432	0.790	0.174	0.019	0.058	0.023	0.339	0.559
Hebei	0.356	0.961	1.395	0.031	0.120	0.078	0.227	0.423	0.924
Heilongjiang	0.454	0.810	1.470	0.122	0.239	0.219	0.141	0.204	0.735
Henan	0.303	0.687	1.351	0.035	0.037	0.078	0.176	0.452	0.892
Hubei	0.345	0.846	1.611	0.034	0.060	0.089	0.243	0.537	1.151
Inner Mongolia	0.354	0.714	1.500	0.094	0.129	0.092	0.157	0.284	1.208
Jiangsu	0.407	1.021	2.047	0.044	0.083	0.032	0.215	0.538	1.732
Jiangxi	0.347	0.731	1.250	0.096	0.039	0.058	0.097	0.491	0.904
Jilin	0.354	0.736	1.504	0.076	0.159	0.172	0.121	0.251	0.824
Liaoning	0.554	0.950	1.895	0.072	0.145	0.131	0.322	0.409	1.192
Ningxia	0.340	0.622	0.895	0.117	0.220	0.086	0.086	0.125	0.550
Qinghai	0.350	0.596	1.065	0.160	0.251	0.173	0.055	0.114	0.579
Shaanxi	0.270	0.636	1.074	0.119	0.120	0.133	0.047	0.236	0.592
Shandong	0.410	1.117	1.969	0.031	0.108	0.101	0.235	0.564	1.320
Shanghai	0.653	1.400	2.506	0.246	0.239	0.135	0.110	0.520	1.688
Shanxi	0.354	0.746	1.486	0.052	0.096	0.130	0.171	0.370	0.926
Sichuan	0.525	0.702	1.128	0.245	0.089	0.066	0.060	0.330	0.802
Tianjin	0.527	1.166	2.513	0.138	0.216	0.131	0.161	0.472	1.809
Xinjiang	0.339	0.746	1.070	0.150	0.277	0.173	0.052	0.133	0.517
Yunnan	0.327	0.675	0.874	0.065	0.111	0.060	0.129	0.284	0.628
Zhejiang	0.521	1.296	2.122	0.039	0.042	0.081	0.284	0.943	1.565
By region									
East	0.454	1.090	1.971	0.187	0.210	0.102	0.221	0.615	1.437
Middle	0.339	0.773	1.466	0.064	0.079	0.091	0.200	0.526	1.090
Northeast	0.488	0.867	1.692	0.098	0.193	0.182	0.258	0.345	1.025
West	0.384	0.661	1.034	0.185	0.175	0.124	0.086	0.295	0.680

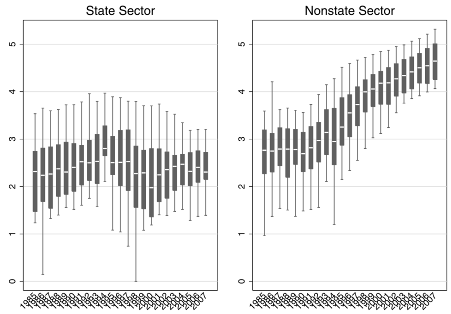

Fig. 1. Box plot of total factor productivity. *Note*: This box plot illustrates, for each year and sector, the log provincial TFP. All values are rescaled relative to the lowest observed value. The dark boxes give the inter-quartile range across provinces for a given year. The median is the white line within each dark box. The bottom and top ends of the thin whisker mark the 5th and 95th percentiles, respectively. The figure illustrates the generally constant TFP in the state sector in all provinces. In the non-state sector, TFP is continuously increasing and the cross-province dispersion is generally declining.

- The big economic point is that China's non-state sector became both much more productive and increasingly dominant for efficient allocation, while the state sector remained relatively unproductive.

5.2 Figure 2: actual versus efficient aggregate TFP

Read:

- The figure plots actual aggregate non-agricultural TFP and the model-implied efficient TFP over time.
- There is a persistent gap throughout the sample, meaning China is always operating inside the efficient frontier.
- The gap narrows until about the mid-1990s and then widens again. In the paper's summary numbers, the aggregate distortion measure is about 0.24 in 1985, 0.18 in 1997, and 0.23 in 2007.
- This is the headline picture of the paper: reform initially reduced misallocation, but the improvement was not sustained.

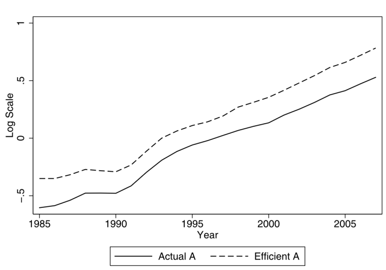

Fig. 2. Productivity over time. *Note*: This plots the observed aggregate non-agricultural TFP in China over time with the model-implied efficient TFP. The increasing gap between the two lines illustrates the aggregate effect of the distortions.

Table 2
Distortions and TFP growth over time, aggregate.

Period	1985–2007	1985–1997	1997–2007
Average distortion	0.202	0.195	0.209
Average efficient TFP growth	6.46%	6.44%	6.49%
Average actual TFP growth	6.52%	6.95%	5.99%
Impact of distortions: actual − efficient	0.06%	0.52%	−0.50%

Table 3
Distortions and TFP growth over time, by region.

Region	Average within province distortion	Average actual TFP growth	TFP growth differential: actual − efficient		
	1985–2007	1985–2007	1985–2007	1985–1997	1997–2007
East	0.087	6.60%	−0.03%	0.36%	−0.51%
Middle	0.145	6.57%	0.24%	0.89%	−0.54%
Northeast	0.139	5.83%	0.07%	0.56%	−0.52%
West	0.158	4.97%	−0.17%	0.41%	−0.85%

Table 4
Robustness: Impact of distortions.

Scenario	Average distortion	TFP growth differential: actual − efficient		
	1985–2007	1985–2007	1985–1997	1997–2007
Baseline	0.20	0.06%	0.52%	−0.50%
$\sigma^{-1} = 3$	0.26	−0.06%	0.39%	−0.60%
$\phi^{-1} = 3$	0.21	0.14%	0.63%	−0.46%
$a = 0.5$	0.23	0.21%	0.96%	−0.69%
Equal w_i	0.26	−0.06%	0.37%	−0.58%

5.3 Table 2: aggregate distortions and TFP growth over time

Read:

- Over the full 1985–2007 period, the average distortion is 0.202, implying actual non-agricultural TFP is about 20% below the efficient level on average.
- Efficient TFP grows at roughly 6.46% per year, while actual TFP grows at 6.52% per year over the full sample. That near equality masks a sharp reversal across subperiods.
- From 1985 to 1997, actual TFP growth exceeds efficient TFP growth by 0.52 percentage points per year, meaning falling distortions contributed positively to growth.
- From 1997 to 2007, actual TFP growth is 0.50 percentage points *below* efficient TFP growth, meaning rising distortions shaved about half a point off annual TFP growth.

5.4 Table 3: regional within-province distortions and growth

Read:

- The East has the lowest average within-province distortion (0.087) and the highest average actual TFP growth (6.60%).
- The West has the highest average within-province distortion (0.158) and the lowest actual TFP growth (4.97%).
- All four regions show the same time pattern as the national data: distortion falls in 1985–1997 and then rises in 1997–2007.
- The post-1997 increase in distortion is most damaging in the West, where it reduces potential TFP growth by about 0.85 percentage points per year over 1997–2007.

5.5 Table 4: robustness to alternative parameter choices

Read:

- The paper varies the elasticity of substitution across provinces, the elasticity across sectors, the labor share, and the province weights.
- Higher substitution elasticities or lower labor shares mechanically raise the level of measured distortions, as expected. For example, the average distortion rises from 0.20 in the baseline to 0.26 when $\sigma^{-1} = 3$ or when province weights are set equal.
- The important point is that the *time pattern* is stable across all specifications: distortions improve before 1997 and worsen after 1997.
- So the core historical conclusion is not driven by one calibration choice.

5.6 Figure 3: overall, within-province, and between-province distortions

Read:

- Panel (a) shows that removing between-province distortions leaves the time pattern of aggregate distortion largely intact, while removing within-province distortions makes the residual distortion roughly flat over time.

- This means the *time variation* in overall misallocation comes mainly from within-province distortions, not from between-province distortions.
- Panel (b) shows that within-province distortions are driven overwhelmingly by capital-market wedges between the state and non-state sectors. Labor-market wedges within provinces are small and relatively stable.
- Panel (c) shows that between-province distortions are driven mainly by labor-market wedges, while between-province capital and product-market wedges are small and declining.

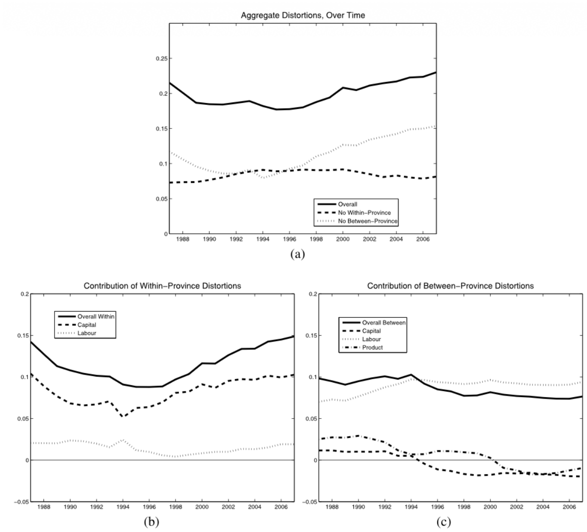

Fig. 3. Distortions within and between provinces of China, 1985–2007. *Note*: Panel (a) illustrates the aggregate non-agricultural TFP loss from overall distortions, the TFP loss from within-province distortions, and the TFP loss from between-province distortions. Panel (b) decomposes the within-province distortions into capital and labour market distortions. Panel (c) decomposes the between-province distortions into capital, labour, and product market distortions.

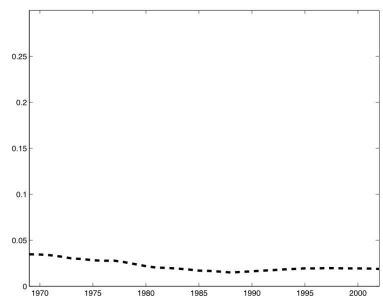

Fig. 4. Between-state distortions through time for the United States. *Note*: This figure uses state-level data from the US to infer the TFP loss from between-province (or state in this case) distortions in labour and capital markets. For comparison with China, see the dashed line in Fig. 3(a).

5.7 Figure 4: US between-state distortions as a benchmark

Read:

- Using a comparable state-level exercise for the United States, the paper finds that between-state distortions lower US productivity by only about 2%, with little time variation.
- The corresponding between-province loss in China is close to 10%, much larger than in the United States.
- This benchmark helps calibrate magnitudes: China's spatial factor-market distortions are not just present; they are large by international standards.

5.8 Figure 5: cross-province dispersion in TFP

Read:

- The figure plots the variance of log TFP across provinces over time and shows a rising trend, especially from the mid-1990s onward.
- This is the key explanation for why between-province labor distortions do not fall despite massive migration. As TFP gaps widen, labor must move more just to keep return differentials from

rising.

- The paper's interpretation is that labor mobility increased, but not fast enough to offset the growing cross-province dispersion in productivity.

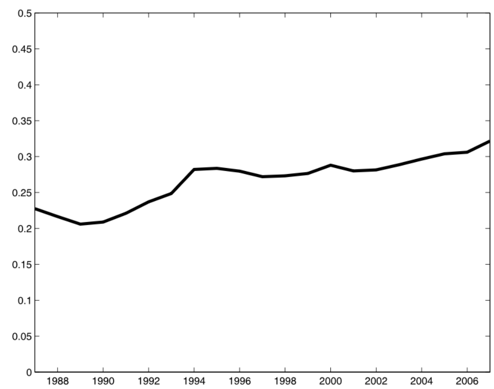

Fig. 5. Cross-province dispersion in TFP. *Note*: This figure plots the variance in log TFP across provinces over time.

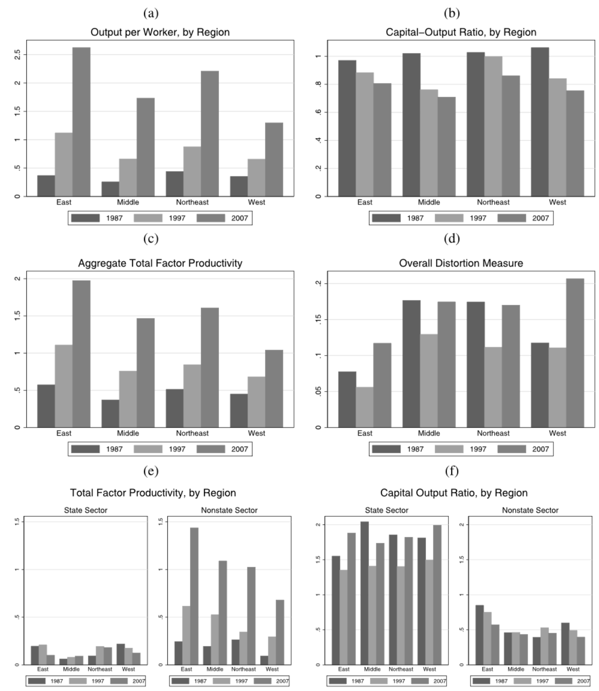

Fig. 6. Comparison by region and sector in China.

Fig. 6(f)). Table 3 shows the average impact of the increased within-province misallocation of capital on provincial TFP growth for the four regions during the period of 1997–2007. It is negative for all four regions. However, within-province misallocation of capital had the largest negative impact on TFP growth in the Western region, reducing potential TFP growth rate by 0.87% a year, and the smallest impact in the Eastern region, reducing potential TFP growth rate by 0.51% a year.

It is also important to note that prior to the mid-1990s the within-province allocation of capital was improving, with the state sector's capital intensity declining from 1987 to 1997 in all four regions. Brandt and Zhu (2000) provide a discussion

5.9 Figure 6: region and sector comparisons, and the Great West policy

Read:

- The West has the lowest output per worker but the highest capital-output ratio, so its weak performance is not a simple story of capital scarcity.
- The deeper problem is low TFP, especially in the state sector. The regional-development push after the late 1990s raised the West's capital-output ratio further without closing the productivity gap relative to the East.
- The paper argues that the "Develop the Great West" policy channeled too much capital into low-TFP state firms, worsening within-province capital misallocation.
- The regional cost was large: the increased within-province capital misallocation reduced potential TFP growth by about 0.87 percentage points per year in the West during 1997–2007, compared with about 0.51 points in the East.

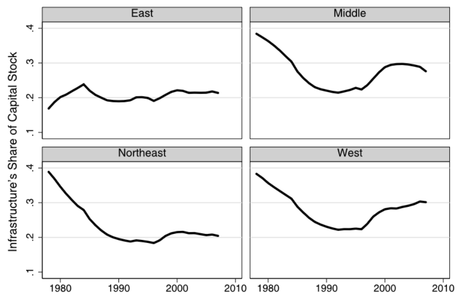

Fig. 7. Infrastructure's share of capital stock.

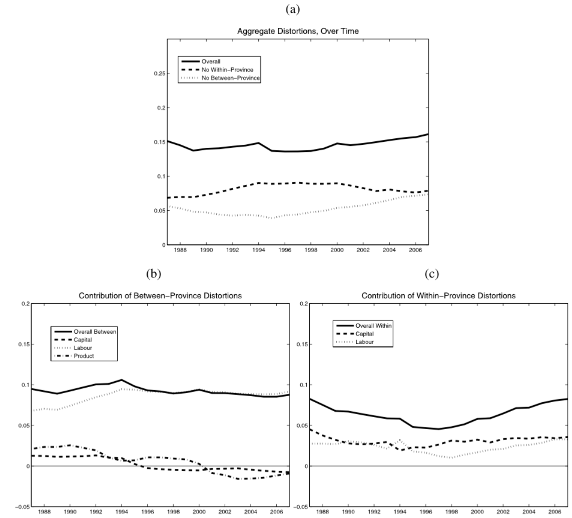

Fig. 8. Distortions between and within provinces of China, infrastructure case. *Note:* These plots are similar to our baseline but with infrastructure capital explicitly accounted for. Panel (a) illustrates the aggregate non-agricultural TFP loss from overall distortions, the TFP loss from within-province distortions, and the TFP loss from between-province distortions. Panel (b) decomposes the within-province distortions into capital and labour market distortions. Panel (c) decomposes the between-province distortions into capital, labour, and product market distortions.

Table 5
Labour shares of output, by sector.

	Non-state	State
1985	56.21%	59.94%
1990	55.70%	60.27%
1995	57.07%	60.40%
2000	56.30%	65.05%
2007	53.38%	68.13%

5.10 Figures 7 and 8: infrastructure-capital robustness

Read:

- Figure 7 shows that the West has the highest infrastructure share of capital and that this share rises again after the mid-1990s, coinciding with the regional investment push.
- Figure 8 recomputes the decomposition with infrastructure explicitly modeled as a separate input.
- Accounting for infrastructure lowers the measured contribution of non-infrastructure capital distortions and raises the role of labor distortions somewhat, but the two central findings survive: between-province labor distortions stay high and stable, and within-province capital distortions remain large and rise after the mid-1990s.

5.11 Table 5: labor shares and industry composition

Read:

- The implied labor share of the state sector rises from about 59.94% in 1985 to 68.13% in 2007, while the non-state labor share stays roughly in the 53–57% range.
- This means the state sector has become more labor intensive, not more capital intensive.
- Therefore, the paper's finding of rising capital misallocation cannot be dismissed as an artifact of the state sector shifting toward capital-heavy industries.

5.12 Overall empirical lesson from the tables and figures

Read:

- The visual evidence lines up tightly with the decomposition results: the within-province state/non-state capital wedge is the moving piece in the aggregate story.
- Spatial labor distortions remain a persistent drag, but they are not the reason the aggregate distortion worsens after the mid-1990s.
- The worsening post-1997 pattern is tied to policy choices that favored state-sector investment, especially in already low-productivity regions.

6 Short summary

- **Conclusion.** China's non-agricultural economy suffered large and persistent productivity losses from factor-market distortions over 1985–2007.
- **Identification insight.** The paper combines a structural allocation model with province-sector data and provincial prices to identify labor, capital, and output wedges up to scale and map them into TFP losses.
- **Empirical lesson.** The big post-1997 deterioration comes from worsening capital misallocation between state and non-state sectors within provinces, while between-province labor distortions remain an important but relatively stable drag on productivity.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Aggregate non-agricultural TFP in China is substantially below the efficient benchmark because labor and capital are misallocated across provinces and across ownership sectors.
- **Quantitative contribution.** The average loss is about 20% over 1985–2007, and by 2007 the distortions lower aggregate non-agricultural TFP by at least a quarter.
- **Historical contribution.** Distortions do not move monotonically with reform. They fall during the first phase of reform and then rise again after 1997.
- **Decomposition contribution.** The worsening after 1997 is almost entirely a within-province capital-market story, not a between-province story.

7.2 Economic intuition

- Efficient allocation sends both labor and capital toward high-TFP province-sector cells.
- China's state sector has systematically lower TFP than the non-state sector, so any policy that channels capital toward the state sector creates a large productivity cost.

- The post-1997 policy push toward western development did increase investment in poorer regions, but much of that investment went through low-productivity state firms. That raised the capital-output ratio without fixing the underlying TFP gap.
- Meanwhile, labor mobility across provinces improved, but the rise in cross-province TFP dispersion was so strong that labor-market distortions remained large even in the face of massive migration.

7.3 Takeaway

This paper is best read as a structural decomposition of Chinese misallocation into *where* it happens and *which wedge* causes it. Its main message is not just that distortions are large, but that the politically important margin is within-province capital allocation between the state and the non-state sectors. If future reform reduces the favored treatment of state firms and allows capital to flow more freely toward higher-productivity private activity, the implied productivity gains are large.